Understanding the Impact of Socioeconomic Factors on Diabetes Prevalence

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Abstract: The legacies of colonialism, imperialism and urbanization have had significant consequences on the lives of a great multitude of human beings. With the concentration of resources and power in the hands of the few, the majority are often forced into exposure to a variety of stimuli that render them vulnerable to disease, poor quality of life, and other difficulties exacerbated by a lack of access to adequate healthcare, nutrition and opportunities for social mobility. Moreover, many human groups have predicated their governance systems on the perpetuation of inequality, ensuring a permanent stratification of resources and poverty to maintain control over their citizens. As a result, those with the opportunity to engage with research, define policies, and most importantly, those with the most power for change often come from more privileged castes. Because of this, when humans look to improve the lives of those who suffer, there remains the risk of missing social or societal attributes that influence the severity or occurrence of disease. Here, we contend that social vulnerability may play just as important of a role in predicting disease outcome as traditional physiological metrics. And in untangling the relationship between social vulnerability and disease outcome, we hope to inform on routes for treatment that go beyond basic medical intervention. To do this we look to the system of diabetes prevalence using machine learning to model disease outcome through our computed SVI (modeled after CDC standard) fed into XGBoost and General Additive Modeling. Our goal is to determine the effectiveness of creating a consolidated single metric (SVI) for determining the effect of socioeconomic factors on health outcomes in complex models.

We maintain that improving the quality of life for all members of a society, improving access to quality food resources, and education, reducing poverty, all stand poised to reduce the prevalence/burden of disease across all levels of class stratification. As researchers we have the unique opportunity and training to step back and look at the systems we study to best understand what drives the trends we see, and in the end we must realize that the answer is deeper than the individual, deeper than the flesh. Therefore, in accordance with what we have seen in the historical record, so long as there is inequality in our world perpetuated by those who hold the reigns of power, it falls to the people to deliver themselves.

Background: Diabetes is a chronic metabolic condition characterized by impaired regulation of blood glucose levels and is associated with significant morbidity and mortality (**American Diabetes Association, 2023**). In the United States, diabetes represents a major public health burden, contributing to cardiovascular disease, kidney failure, vision loss, and reduced quality of life. As a result, identifying individuals at increased risk for diabetes is an important goal for prevention and early intervention. Diabetes is a heterogeneous disease that primarily includes Type 1 and Type 2 diabetes. Type 1 diabetes is an autoimmune condition resulting from destruction of pancreatic beta cells and typically presents earlier in life. In contrast, Type 2 diabetes accounts for the majority of cases and is strongly associated with both modifiable and non-modifiable risk factors, including obesity, physical inactivity, diet, age, and genetic

predisposition (**American Diabetes Association, 2023**). Because Type 2 diabetes develops gradually and is influenced by environmental and behavioral factors, it is particularly well suited for risk modeling using population-level data. In this study, we analyze data from the Centers for Disease Control and Prevention (CDC) Behavioral Risk Factor Surveillance System, a large survey-based dataset that includes responses from over 250,000 individuals. The dataset contains a broad range of demographic, lifestyle, social, and clinical variables, along with information on diabetes status. For this analysis, the original outcome variable distinguishing between no diabetes, prediabetes, and diabetes was simplified into a binary classification task by grouping individuals with diabetes or prediabetes into a single positive class and individuals without diabetes into a negative class.

While clinical predictors such as age and body mass index are well-established risk factors for diabetes, a growing body of research highlights the role of social determinants of health in shaping diabetes risk and outcomes. Socioeconomic status, educational attainment, and access to healthcare have been consistently associated with diabetes prevalence, disease management, and diabetes-related complications (**Hill-Briggs et al., 2021**). Individuals with lower income or limited access to healthcare may face barriers to preventive care, timely diagnosis, and effective disease management, increasing overall vulnerability. Based on this understanding, we expected that individuals' social vulnerability would be a useful predictor of disease state. We further hypothesized that several related socioeconomic variables could be combined into a single Social Vulnerability Index (SVI) without losing meaningful predictive information. Variables such as income level, educational attainment, health insurance coverage, and the inability to afford medical care are often correlated and may reflect a shared underlying dimension of vulnerability. By summarizing these factors into a single metric, we expected the SVI to capture this latent construct while reducing redundancy among predictors.

Prior research has consistently demonstrated that social determinants of health play a critical role in diabetes risk and disease progression. Lower socioeconomic status, reduced educational attainment, and limited access to healthcare have been associated with higher prevalence of Type 2 diabetes, poorer glycemic control, and increased risk of diabetes-related complications. These findings support the inclusion of social and economic factors in predictive modeling of diabetes risk and motivate the development of composite measures such as a Social Vulnerability Index (**Hill-Briggs et al., 2021**).

Finally, we reasoned that different modeling approaches would be needed to capture these relationships in complementary ways. A Generalized Additive Model (GAM) was expected to provide interpretable nonlinear effects that clarify how individual social and clinical factors relate to diabetes risk. In contrast, a more flexible tree-based method such as XGBoost was expected to better capture complex interactions and non-additive patterns in the data. Together, these models were selected to balance interpretability with predictive performance.

Methods

Imputation: Missingness was first quantified across all variables to assess its extent and structure. Approximately 10% of observations were missing per variable on average. We next examined the distribution of missingness at the row level, identifying the number of records with varying proportions of missing values. Only 189 out of approximately 250,000 observations (<0.05%) were missing more than 35% of variables, a proportion deemed too small to materially influence downstream analyses.To evaluate whether missingness occurred randomly, Kolmogorov–Smirnov (KS) tests were used to compare the

distributions of observed and missing values across variables and no outliers were identified. Based on these diagnostics, multiple imputation strategies were considered to ensure robustness, including Fast Random Forest Imputation and Multiple Imputation by Chained Equations (MICE). All subsequent analyses were conducted on MICE imputed dataset. We performed an 80/20% train/test split of the imputed data.

> **MICE** is a method of imputation that starts by replacing all the missing values for a given column with an initial approximation, such as the mean or median. As it imputes values for an individual column with missing data, MICE builds a predictive model that leverages the relationships between the variables to build different models (linear, logistic) to create imputed datasets. The model imputes the missing values for a single column at a time, repeating this process of updating the approximations for other columns iteratively. This chain process continues until MICE computes equally plausible, complete datasets. Then, statistical analysis is generally performed with each imputed dataset, before the results are pooled to produce a more robust metric that can account for the uncertainty inherent to the imputation process.

Exploratory Data Analysis: Exploratory analyses were conducted to characterize distributions and associations among variables prior to index construction and modeling. Pairwise correlations and summary statistics were examined to identify collinearity and general trends among predictors. To ensure consistent interpretation of healthcare access variables, binary indicators related to cost-based barriers were reverse-coded such that higher values consistently reflected greater access. Exploratory analyses were performed in R using the dplyr, tidyr, and ggplot2 packages.

SVI INDEX CREATION:

Variable selection and preprocessing: The Social Vulnerability Index (SVI) was constructed using education, income, healthcare coverage, and avoidance of medical care due to cost. Ordinal socioeconomic variables (education and income) were treated as quantitative inputs, while healthcare access variables were treated as categorical. To accommodate potential nonlinear relationships in the ordinal socioeconomic variables, penalized regression spline basis expansions were generated for education and income using the mgcv framework. Spline basis columns with zero variance were removed to prevent numerical instability. Remaining basis columns were assembled into a quantitative feature matrix.

Mixed-data principal component analysis: A mixed-data principal component analysis (PCA) was performed using PCAmix, combining the spline-expanded quantitative matrix with categorical healthcare access variables. The first principal component (PC1) was extracted for each individual and defined as the Social Vulnerability Index (SVI). The resulting index was rescaled to the unit interval (0–1) using min–max normalization to facilitate interpretability and comparability across models.

Interaction-augmented SVI variants: To assess whether interactions among socioeconomic and healthcare access variables altered the structure of latent vulnerability, additional SVI variants were constructed. First, pairwise multiplicative interactions between spline basis

functions for education and income were added to the quantitative matrix prior to mixed PCA. PC1 was re-extracted and rescaled to produce an interaction-aware SVI. A final, fully interaction-augmented SVI was constructed by incorporating higher-order interaction terms, including spline-by-spline interactions for all quantitative pairs, spline-by-categorical interactions, categorical-by-categorical interactions, and selected three- and four-way interactions. These interaction terms were appended to the spline main effects, and mixed PCA was repeated. The first principal component from this expanded feature space was extracted and rescaled to generate the final SVI with all interactions.

Modeling Diabetes Prevalence Using SVI: Diabetes status was defined using a binary indicator. To examine the relationship between social vulnerability and diabetes prevalence, the scaled SVI was visualized using histograms with density overlays, boxplots stratified by diabetes status, and outcome-stratified density plots. Participants were further grouped into quartiles of SVI, and mean diabetes prevalence within each quartile was computed to assess monotonic trends.An initial generalized additive model (GAM) was used to model diabetes probability as a function of SVI. Predicted probabilities across a grid of SVI values were used to generate smooth effect plots with 95% confidence intervals for visualization purposes.

Generalized Additive Models (GAMs and BAMs):

Model specification: Generalized additive models were fit using a binomial family and logit link to flexibly model nonlinear associations between covariates and diabetes risk. The SVI was included as a linear term to preserve interpretability of the vulnerability gradient, while allowing nonlinear effects to be captured by other covariates. Continuous covariates were modeled using penalized splines, and categorical covariates were included as parametric factors.

Large-scale GAMs with interactions: To efficiently accommodate nonlinearities and high-dimensional interaction structures, large-scale GAMs were fit using the bam() function from the mgcv package. Models incorporated spline main effects for continuous variables, parametric main effects for categorical variables, pairwise tensor-product spline interactions between continuous covariates, and continuous interactions by category using factor-specific smooths. Shrinkage penalties were applied to automatically downweight uninformative terms. Models were fit using discrete approximations and parallel computation to improve scalability.

Model evaluation and interpretation: Predicted probabilities from fitted GAM models were evaluated using receiver operating characteristic (ROC) curves and area under the curve (AUC). Calibration was assessed by grouping predicted probabilities into bins and comparing mean predicted risk with observed diabetes prevalence. Classification performance was further evaluated across multiple probability thresholds (0.1–0.5) using confusion matrices and derived metrics including accuracy, sensitivity, specificity, precision, recall, and F1 score. Effective degrees of freedom (EDF) were extracted from model summaries, and terms with EDF greater than 0.1 were considered to contribute meaningfully to model fit.

Gradient Boosted Tree Models (XGBoost): Gradient boosted decision tree models were implemented using the XGBoost framework as a complementary, nonparametric approach to modeling diabetes risk. These models mirrored the GAM models in working with both the SVI metric and the individual socioeconomic variables so that the results were able to be compared, and the performance of the SVI assessed. Models were fit with a binary logistic objective and optimized using cross-validation to maximize AUC. Three predictor configurations were evaluated: (i) linear model with individual socioeconomic variables (SES-only), (ii) SES-only allowing for pairwise interactions, (iii) SES-only allowing for pairwise interactions and feature selection, (iv) composite SVI only (SVI-only) allowing for only linear terms, (v) SVI-only allowing for nonlinear terms. No interaction (linear) models were created by setting max_depth to 1, while models with pairwise interactions were created by setting max_depth to 2. All models were subject to 5-fold cross validation to tune hyperparameters such as minimum loss reduction, learning rate, and minimum child weight. Feature selection was performed on the SES-only model by setting colsample_bytree equal to c(0.6, 0.8), allowing for selection of some variables but not others and improvement of the model when using variables with real effect. Shallow trees were favored to limit overfitting. Feature importance measures were extracted for multivariable models to identify predictors contributing most strongly to classification performance. To assess individual predictor contributions, additional permutation importance was computed for each of the SES models by shuffling each variable in the test set while holding the other variables constant and quantifying the resulting impact on the model's AUC.

Results: Preliminary analysis of the data showed that only about 13% of our population falls into the class "diabetic". This is marginally higher than the reported prevalence of diabetes in the American populace which sits just above 11% rising to 15% when we account for the undiagnosed proportion (CDC 2021). This suggests our data accurately reflects the natural prevalence of diabetes in the American public. Additionally looking at the comparison of PC1 vs PC2 we see that there is very minor clustering but not enough to fully pull apart the diabetic (coded as 0) vs non diabetic cases (Figure 4). However we see general grouping with fair overlap concentrating the highest density of diabetic observations on the right side (higher scores of PC1). This gradient of separation is what we based our index for social vulnerability on.

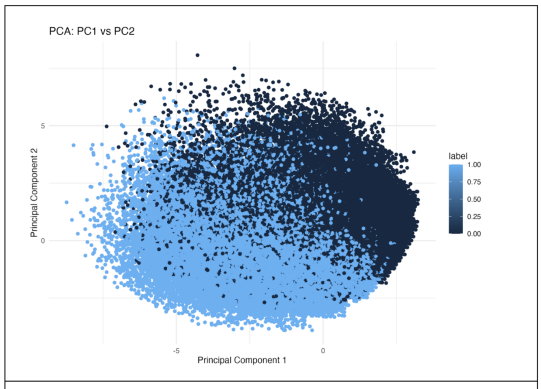
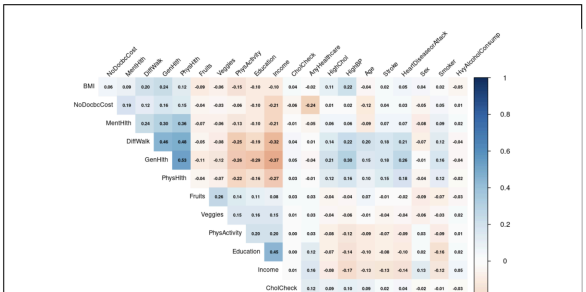


Figure 4: PC1 vs PC2 SVI

Prior to constructing the Social Vulnerability index we created a correlation matrix to look for correlations between our parameters (Figure 1). The strongest positive correlations were unsurprisingly between general and physical health, between physical



and general health and difficulty walking. But the correlation strength was only a max of 0.48. It should also be noted that general health reports followed High Blood pressure more closely than they followed high cholesterol.

Another interesting correlation was that of education and income pointing towards the idea that income-social stratification often coincides with less effective educational infrastructure and subsequently lower levels of education. Finally other positive trends were observed that fall along the lines of physiology, namely that factors like high blood pressure correlated ~0.3 with high cholesterol and age. As for negative correlations, income was negatively correlated with several factors often associated with vulnerability, namely physical health, general health, difficulty walking, mental health, doctoral avoidance on account of prohibitive costs. This suggests that merging these terms into a single metric could effectively reduce the dimensionality of the data and better tease apart complex relationships like disease states. SVI, which contains social variables including income, is a common metric used to explain vulnerability to disease; we examined the potential explanatory power of this metric.

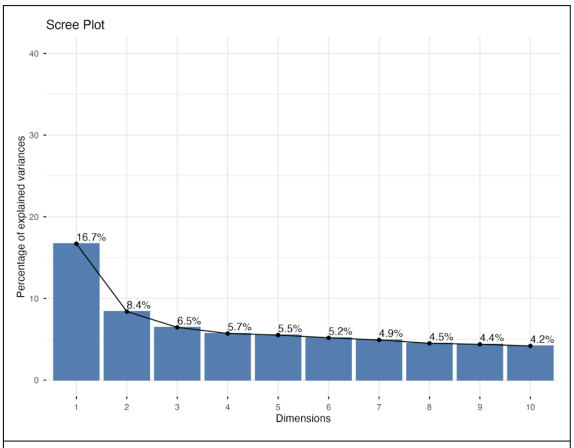


Figure 2: Scree plot showing variance explained by each principal component.

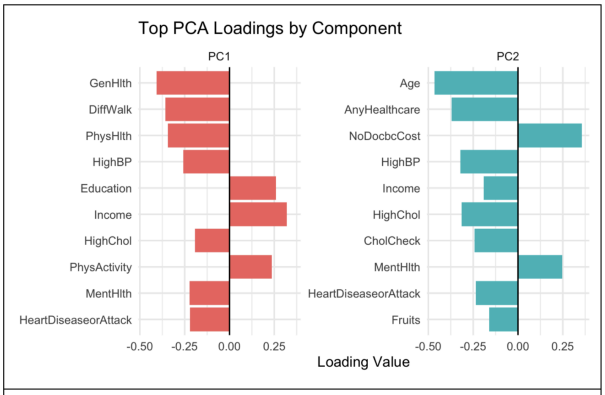


Figure 3: Loadings for PC1 and PC2

We performed PCA, to determine if our dataset was linearly separable and whether it could be handled using linear methods. Our PC1 explained 16.7% of the variance; PC2 explained around 8.4% of the variance (Figure 2). Looking at the actual loadings, we see that many variables contribute to PC1 and PC2, and that some variables contribute strongly to both (Figure 3). In addition, the variance of selected variables observed on PC1 and PC2 make it clear that there is significantly nonlinearity and collinearity to be addressed in the dataset. It is also clear from these results that the data is not linearly separable. For these reasons, we use splines to address the nonlinearity that we observe. We also attempt to reduce our potentially collinear variables through the use of SVI (and later through feature selection).

For SVI, we created our index from four variables linked to social vulnerability: Income, Education, Healthcare Coverage, and Avoidance of Medical Care due to Cost. The variance of these four variables was explained via principal components, using penalized spline basis expansions for the two continuous variables, Income and Education. PCA was performed with no interactions, in addition to with only an interaction between income and education. These approaches led to a PC1 explaining 47.66% and 43.92% or variance between the four variables. All possible interactions between the four variables were

used in the final PCA, leading to a PC1 which explained 59.12% of the variance of the training data set (Figure 6). As expected, those with diabetes had a higher SVI score (more vulnerable) (Figure 7). Vulnerability as measured by SVI was not normally distributed (Figure 8).

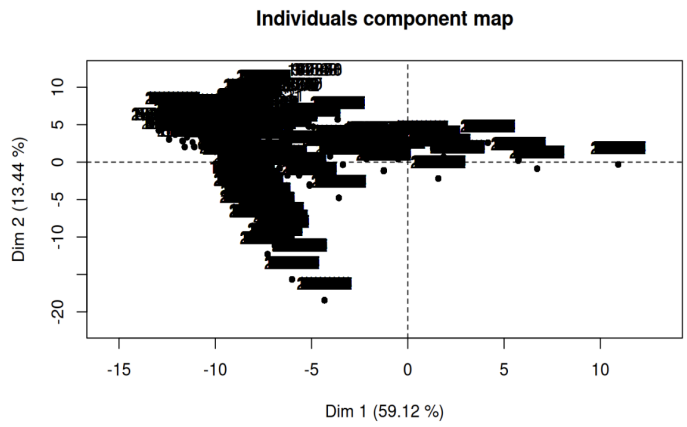


Figure 6: SVI Individual Component Map showing the placement of individuals on the PC1 and PC2 axis after accounting for all possible interactions between the four SVI variables.

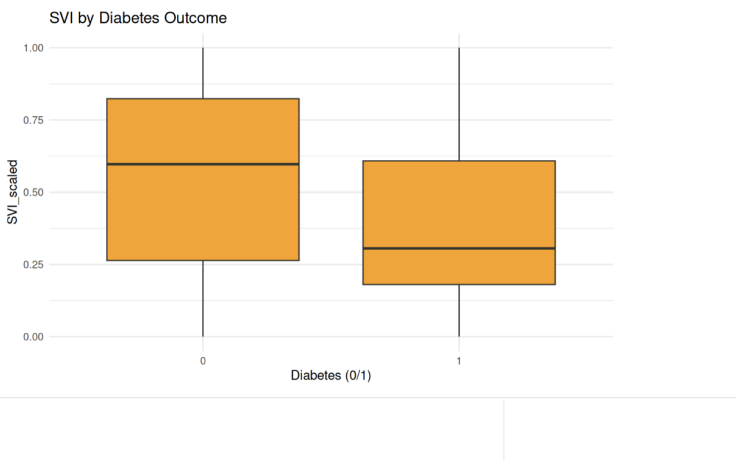


Figure 7: Box plot showing the scaled SVI score of those with and without diabetes.

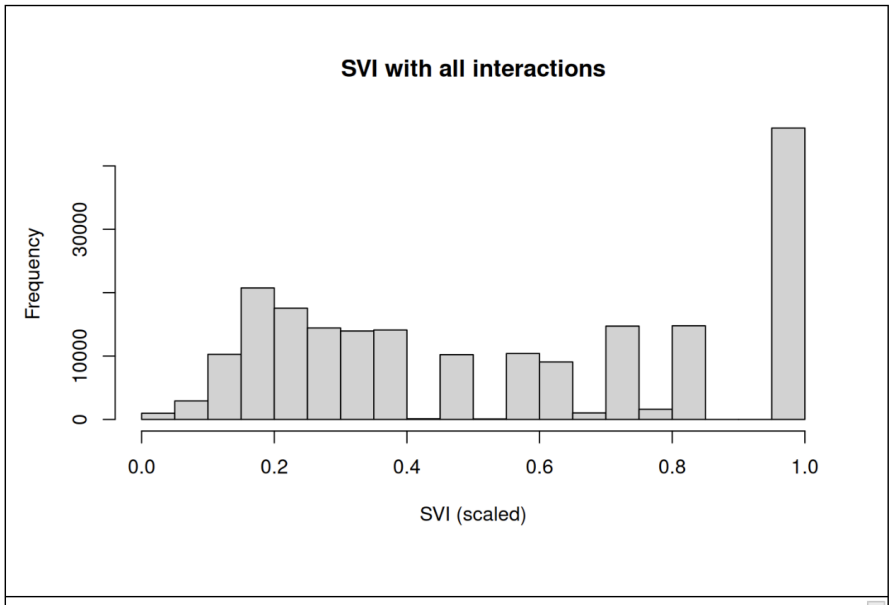


Figure 8: The distribution of SVI among individuals in the train dataset. Most vulnerable is 0 and least vulnerable is 1.

GAM: A look at our ROC curves shows strong model performance in both test and training sets with our AUC being around 0.81 for every case (Figure 9). "SVI" plots use the SVI metric, while "SE Sep" plots use the four individual socioeconomic variables. While all of the curves look relatively similar, the GAM model using SVI with no shrinkage and interactions included was the best performing of all the models for the test data. It also was highly interpretable, as examining the parametric coefficients of the model clearly indicates (Figure 10). SVI was a highly important parameter, comparable with several of the clinical variables like Previous Heart Attack and Previous Stroke. Based on the estimate and the log-odds ratio of SVI, going from 0 to 1 in SVI (from most to least vulnerable), decreases likelihood of diabetes by 29.88%. This is a far more interpretable result than the one produced by the model (with the same setup regarding interactions and shrinkage) using individual socioeconomic variables (Figure 11). For this model, the effect of these variables is spread through interactions with many other variables, making the effect observed from the parametric coefficients (and attributable to one metric) appear weaker (or non-existent in the case of income and education) by comparison.

```
Parametric coefficients:
                      Estimate Std. Error z value Pr(>|z|)
(Intercept)           -3.34041    0.28555 -11.698  < 2e-16 ***
SVI_scaled            -0.35510    0.05285  -6.718 1.84e-11 ***
HighBP1                0.76130    0.06920  11.002  < 2e-16 ***
HighChol1              0.81510    0.06835  11.926  < 2e-16 ***
CholCheck1             0.87312    0.24007   3.637 0.000276 ***
Smoker1               -0.16156    0.11756  -1.374 0.169352
Stroke1                0.60708    0.14089   4.309 1.64e-05 ***
HeartDiseaseorAttack1  0.37694    0.07504   5.024 5.07e-07 ***
PhysActivity1         -0.09921    0.04102  -2.418 0.015587 *
Fruits1               -0.03986    0.03297  -1.209 0.226689
Veggies1              -0.03200    0.07258  -0.441 0.659331
HvyAlcoholConsump1    -0.63053    0.10166  -6.202 5.57e-10 ***
DiffWalk1              0.43085    0.13460   3.201 0.001370 **
Sex1                   0.11215    0.15277   0.734 0.462875
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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Figure 10: Parametric coefficients of the best-performing model (according to AUC), using SVI in the place of four socioeconomic variables, without feature shrinkage, and with pairwise interaction terms between variables.

```
Parametric coefficients:
                      Estimate Std. Error z value Pr(>|z|)
(Intercept)           -3.46641    0.22963 -15.096  < 2e-16 ***
HighBP1                0.84411    0.09513   8.873  < 2e-16 ***
HighChol1              0.68420    0.02944  23.238  < 2e-16 ***
CholCheck1             0.97546    0.15092   6.463 1.02e-10 ***
Smoker1               -0.03124    0.04346  -0.719 0.472198
Stroke1                0.27251    0.05498   4.957 7.17e-07 ***
HeartDiseaseorAttack1  0.50522    0.04370  11.560  < 2e-16 ***
PhysActivity1         -0.09278    0.06076  -1.527 0.126793
Fruits1                0.09111    0.04489   2.030 0.042390 *
Veggies1              -0.01496    0.06108  -0.245 0.806485
HvyAlcoholConsump1    -0.68203    0.19824  -3.440 0.000581 ***
DiffWalk1              0.36846    0.07958   4.630 3.65e-06 ***
Sex1                   0.10829    0.07083   1.529 0.126276
NoDocbcCost_flipped1   0.01665    0.06840   0.243 0.807637
AnyHealthcare1        -0.10821    0.14309  -0.756 0.449507
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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
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Figure 11: Parametric coefficients of the model using individual socioeconomic variables instead of SVI, without feature shrinkage, and with pairwise interaction terms between variables.

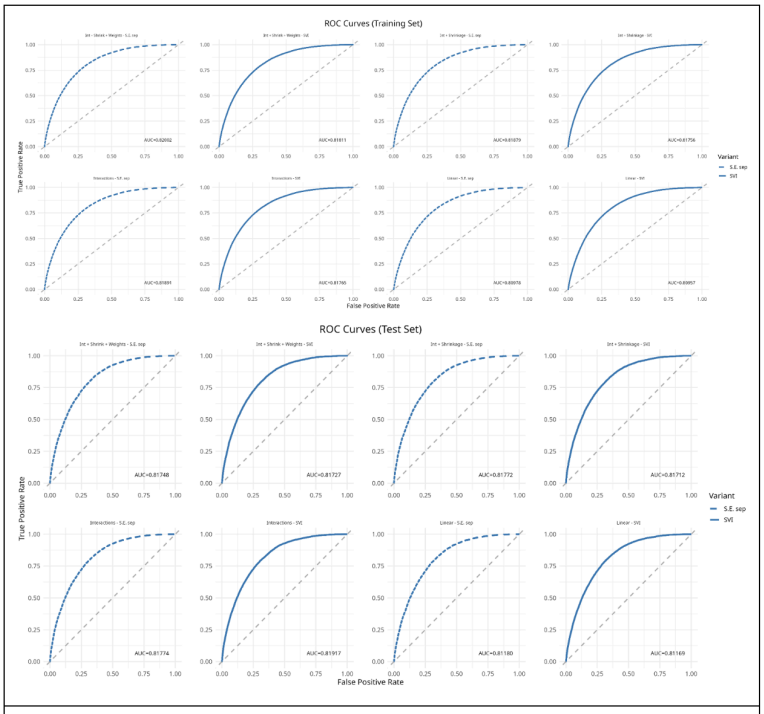

Figure 9: Test and Training Receiver Curves used for Modeling.GAM: with different models (linear only, pairwise interactions, pairwise interactions with feature selection, pairwise interactions with feature selection and weighting for diabetic prevalence)

XGBoost: For XGBoost a set of analyses of models with independent socioeconomic variables and with SVI as the only variables were performed. In this case, while all models were comparable in AUC for the training dataset, the SVI models performed slightly worse than the SES models when evaluated linearly and nonlinearly (Figures 12, 13). However, the XGBoost models had a much lower AUC compared to the GAM models; the GAM models were more predictive of diabetes when evaluated on the test dataset (as expected: the GAM included all variables in some form, while the XGBoost only had socioeconomic variables).

The XGBoost training ROC curves were moderate and similar across all combinations between the nonlinear and linear terms as well as with SES and SVI variables. It demonstrates that there is no significant information gain from altering the model to appreciate interaction terms, feature selection, or nonlinear modeling of the SVI. The greater complexity afforded by the model does not result in increased predictive power.

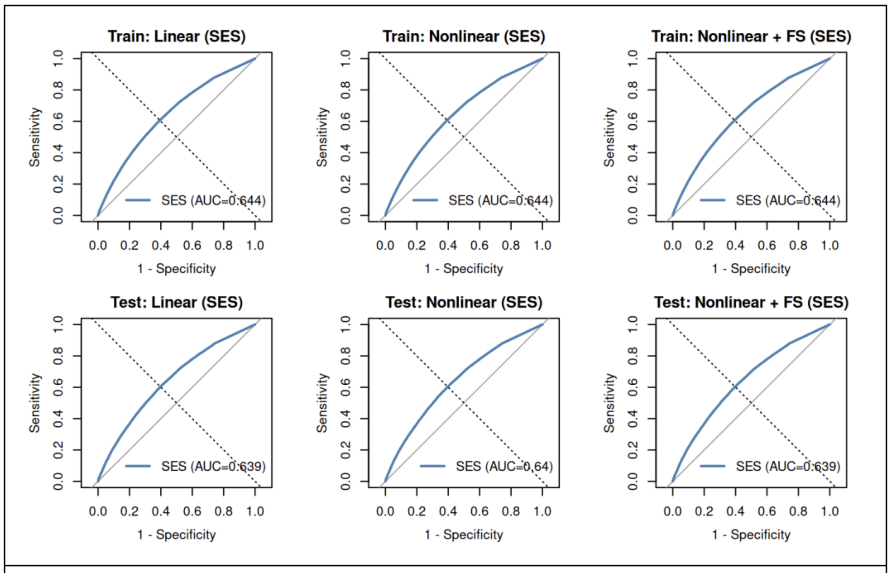

Figure 12: Test and Training XGBoost Receiver Curves used for models using independent socioeconomic variables and different characteristics (linear only, pairwise interactions, pairwise interactions with feature selection)

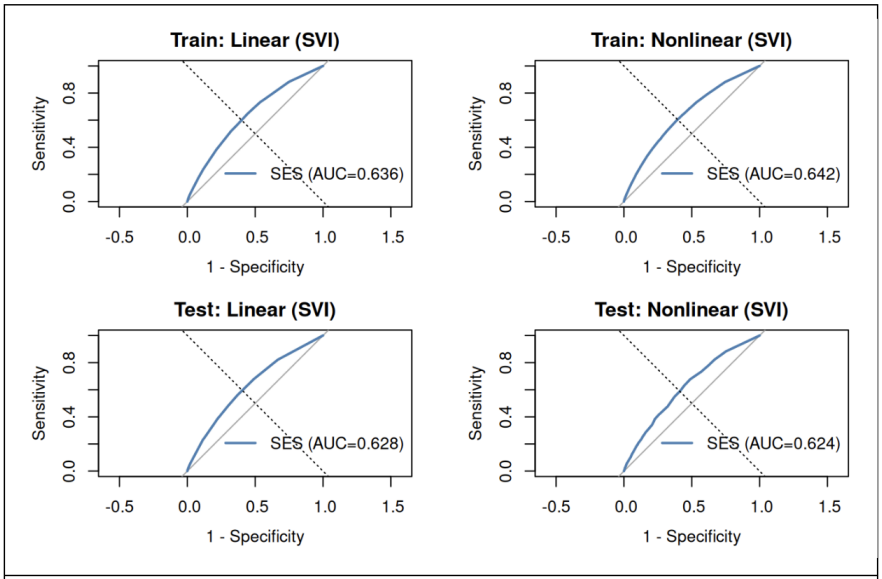

Figure 13: Test and Training XGBoost Receiver Curves used for models using the SVI metric and different characteristics (linear only, pairwise interactions)

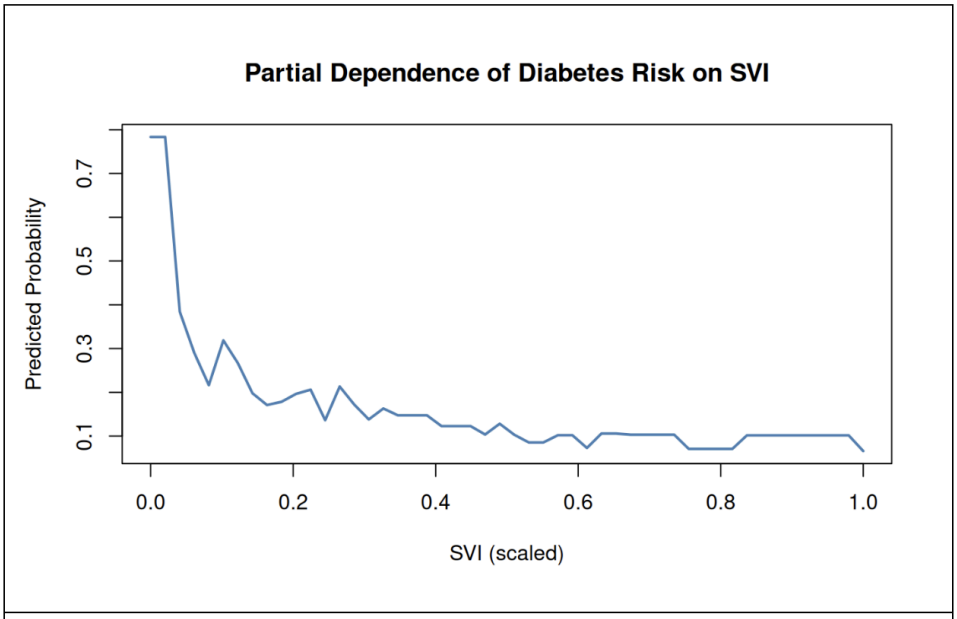

Figure 14: Partial Dependence Plot of scaled SVI against the probability of an individual having diabetes. More vulnerable is 0 and least vulnerable is 1.

The partial independence plot above (Figure 14) shows the relationship between the SVI feature and the predicted probability of diabetes. When considering the nonlinear XGBoost SVI model, we observe that the model assigns a much higher diabetes risk at low SVI values (greater social vulnerability). As the SVI decreases, the model predicts that the risk of diabetes decreases rapidly.

For XGBoost, the SVI-only model required a higher learning rate and fewer boosting rounds to reach optimal performance, reflecting rapid convergence (as expected due to its use of a single parameter) and limited model capacity when using a single composite predictor. With a larger learning rate, each tree applied a stronger correction to the predicted log-odds, allowing the model to have fewer boosting rounds for it to reach its best validation AUC. This shows a limitation of SVI in this kind of tree-based analysis.

Discussion:

In this study, we explored whether social vulnerability contributes meaningfully to diabetes risk and whether this information can be effectively summarized into a single composite metric. Using a large CDC survey dataset, we evaluated a Social Vulnerability Index (SVI) derived from socioeconomic and healthcare access variables and examined its performance across in an interpretable modeling approach and assessed its overall predictive power as a standalone metric in a flexible modeling approach. Overall, the results suggest that social vulnerability captures important information related to diabetes risk and can complement traditional clinical predictors.

Across our interpretable GAM models, well-established clinical variables such as age, body mass index, and measures of physical and general health emerged as strong predictors of diabetes status, consistent with existing biological and epidemiological evidence. This alignment with known risk factors provides confidence that the modeling framework successfully captured meaningful signals in the data.

Importantly, the inclusion of social vulnerability measures did not replace these clinical predictors, but instead added broader contextual information related to diabetes risk.

One of the central findings of this analysis was that the composite SVI carried predictive value beyond that of individual socioeconomic variables considered separately in the GAM. When comparing SVI-based GAM models to GAM models using individual socioeconomic indicators, the SVI demonstrated comparable performance; a GAM model using SVI achieved the highest AUC of any model tested. The resulting model was also highly interpretable, showcasing the effect of social vulnerability on likelihood of diabetes.

This suggests that the SVI captures an underlying dimension of social risk that is not fully represented by individual variables alone, supporting the interpretation of social vulnerability as a latent construct relevant to diabetes outcomes. The comparison between modeling approaches further highlights this result. The GAM offered strong interpretability, allowing smooth and intuitive visualization of nonlinear relationships between predictors and diabetes risk while maintaining stable performance and reasonable calibration. Overall, our GAM models performed comparably for both training and test data, though they were not perfectly predictive of diabetes. The model performed similarly on train and test data, showing their relatively higher bias compared to lower variance.

In contrast, XGBoost provided a more flexible modeling framework capable of capturing complex interactions and non-additive patterns in the data. Perhaps as a result of this, XGBoost models incorporating the SVI achieved slightly lower test data predictive performance comparable to models using multiple individual socioeconomic variables. SVI captures a majority of the variance of the socioeconomic variables, but representing the four socioeconomic variables independently is more predictive when these are the only variables in a model. To better fit the data and model complex interactions sufficiently for high performance, it may be necessary to avoid cumulative metrics. The difference in predictive performance between the SVI and SES models when evaluating the test dataset likely is due to the relatively higher bias of the SVI models. Together, these approaches provided differing but useful perspectives: GAMs clarified how key predictors relate to diabetes risk and showed how a cumulative metric can succeed in a less flexible type of model. XGBoost showed a limitation of this type of metric as a standalone predictor, however it provided insights into how much SVI and similar metrics can measure up to models not simplified in this way. In addition, performance was still comparable for the SVI XGBoost, so for models which are prone to overfitting (and should be simplified to improve performance), SVI has a use.

Several limitations should be acknowledged. The dataset is survey-based and relies on self-reported information, which may introduce reporting bias. In addition, diabetes and prediabetes cases were underrepresented relative to non-diabetic individuals, leading to reduced sensitivity for identifying positive cases despite reasonable overall performance. We attempted weighting of diabetic individuals, leading to a GAM model with comparable performance, though heavily shifted toward sensitivity and away from specificity, compared to a non-weighted GAM model. Although multiple imputation was used to address missing data, imputation introduces additional uncertainty. Finally, while associations between social vulnerability and diabetes risk were observed, these results should not be interpreted as causal based purely on our results..

There are multiple future directions we could take in continuing to analyze this dataset. While weighting such that diabetics had equal representation as non-diabetics did not improve performance significantly while also greatly decreasing the specificity, weighting to a lower extent could potentially improve model performance as evidenced by the AUC. Performing a gradient of weights, checking for improvements in AUC is one way to potentially improve overall model performance. Another future direction would be to fit different models on subsets on the data, based on where an individual falls within SVI. It is possible to confirm whether models are better fitting overall, or have different important variables depending on social vulnerability. For those with different vulnerabilities, different variables related to health may play a role given that access to healthcare and health interventions change.

In summary, these findings suggest that social vulnerability can be meaningfully incorporated into machine learning models of diabetes risk through a composite index. The SVI provides a concise and interpretable representation of social risk factors that complements traditional clinical predictors and performs robustly across modeling approaches. Taken together, the results underscore the importance of considering social determinants of health in predictive modeling and highlight the potential value of integrating social context into approaches aimed at understanding and addressing diabetes risk.

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